Quiz 05. Higher-order, linear ODEs with constant coefficients
prepared by C. Cahig *for* ENS181 (AY 2026-2027, S1)

Instructions. In the following problems, the character d is reserved for derivatives and differentials so that $\mathrm{d}\alpha$ is "the differential of α". As is customary, e is the base of the natural logarithm, and $\exp(abc) = e^{abc}$.

Regard these linear ODEs in the succeeding Scenarios.
1. $32x' - 10x + x^{(5)} + 19x''' - 5x^{(4)} - 37x'' = g_1(y)$
2. $16z^{(4)} - z = g_2(x)$
3. $3y' - 3y'' + y''' - y = g_3(s)$
4. $\ddot{w} + \dot{w} = g_4(t)$

Scenario 1. For each of the ODEs listed above, determine the roots of its auxiliary equation. Report the real roots first, in increasing magnitude, followed by the complex roots, also in increasing magnitude. Express a complex conjugate root pair $x_r \pm jx_i$ as $(x_r;\ x_I)$, that is, a semicolon and space separating x_r and x_i. Do not condense repeating roots in one entry. Separate the roots by a semicolon and a space.

Scenario 2. Take $g_1, g_2, \ldots, g_5 = 0$, and the following initial conditions.
- $x(0) = 2, \quad x'(0) = 2, \quad x''(0) = -7, \quad x'''(0) = -25, \quad x^{(4)}(0) = 29$
- $z(0) = 1, z'(0) = 1, z''(0) = 0.25, z'''(0) = 0$
- $y(0) = 1, y'(0) = 1, y''(0) = 3$
- $w(0) = 2, w'(0) = 0, w''(0) = -1$

Determine the following values.
1. $x(-1.70)$
2. $z(1.19)$
3. $y(-2.14)$
4. $w(0.67)$

Scenario 3. Take
- $g_1(y) = 17$,
- $g_2(x) = e^{0.5x}$,
- $g_3(s) = e^s + 16 - s$,
- $g_4(t) = \tan t$,

and the following boundary conditions
- $x(0) = 0, \quad x'(0) = x''(0) = 1.7, \quad x(1) = 1.7(e-1), \quad x'(1) = 1.7e$
- $z(0) = 0, z'(0) = z''(0) = 0.125, z''(2) = 0.25e$
- $y(0) = 0, y'(0) = 14, y''(1) = \frac{91}{6}e$
- $w(0) = 0, w'(0) = 0, w'(\pi/3) = -0.5\ln\left(2 + \sqrt{3}\right)$

Determine the following values.
1. $x(1.70)$
2. $z(-1.19)$
3. $y(2.14)$
4. $w(-0.67)$

Marking. Each required item above is worth two (2) points. The total percentage rating a student gets for this Activity is computed as $0.59(0.9^m r) + 0.41p$, where m is the number of unmet requirements and violated instructions, r is the raw score (in percent), and p is the peer rating (in percent) by his/her groupmates.

Submission. Answers shall be communicated through the accompanying Google Form. Peer ratings shall be communicated via the submission platform in the Google Classroom.